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Week-1

Instructor: D.MSI

Computer Networks
1.5 hours in a week, 3.00 Cr.

Dr. Mohammad Shahidul Islam
Associate Professor

Reference Book(s):

1. Computer Networks - Andrew S. Tanenbaum
2. Computer Networks: Protocols, Standards, and Interfaces - Uyless Black
3. Internetworking with TCP/IP: Principles, Protocols, Architecture - D. E. Comer
4. TCP/IP Illustrated Vol. I - W. R. Stevens

Course Contents:

- Protocol hierarchies; Data link control: HDLC;
- DLL in Internet; DLL of ATM; LAN Protocols: Standards IEEE 802;
- Hubs, Bridges, and Switches, FDDI,
- Fast Ethernet; Routing Algorithm;
- Internetworking, WAN; Fragmentation; Firewalls;
- IPV4, IPV6, ARP, RARP, Mobile IP,
- Network layer of ATM;
- Transport Protocols; Transmission Control Protocol: Connection Management,

Course Contents:

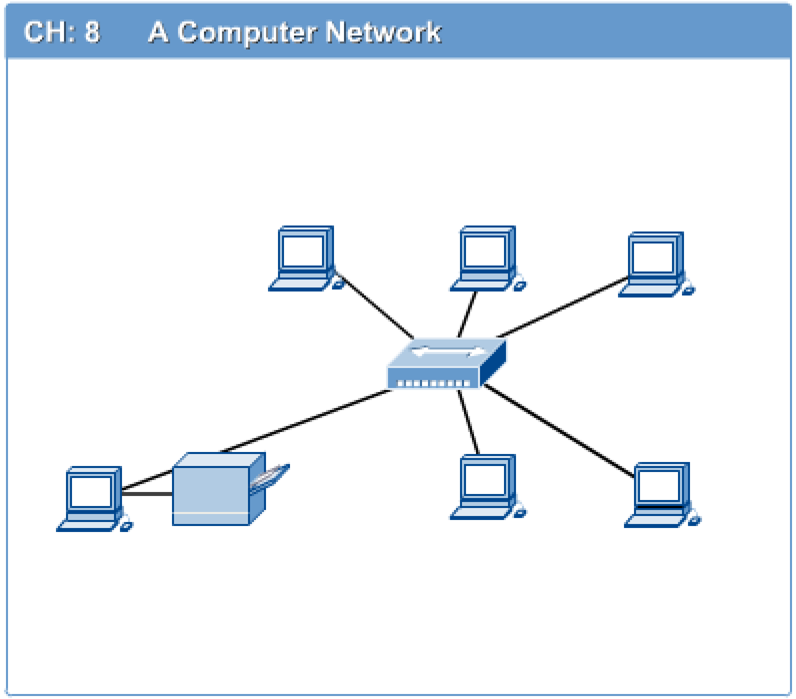
- Transmission Policy, Congestion Control, Timer Management;
- UDP; AAL of ATM; wireless networks,
- mobile computing, and high speed networks;
- Gigabit Ethernet; Domain Name System: Name servers;
- Email and Its privacy; SNMP; HTTP;
- World Wide Web; Network security: Cryptography,
- DES, IDEA, public key algorithm;
- Authentication; Digital signatures.

Today

- Computer Network basics
- Protocol hierarchies;
- Data link control: HDLC;

Computer Networks

- Define?
- Computer network connects two or more autonomous computers.
- The computers can be geographically located anywhere.



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LAN, MAN & WAN

- Network in small geographical Area (Room, Building or a Campus) is called LAN (Local Area Network)
- Network in a City is call MAN (Metropolitan Area Network)
- Network spread geographically (Country or across Globe) is called WAN (Wide Area Network)

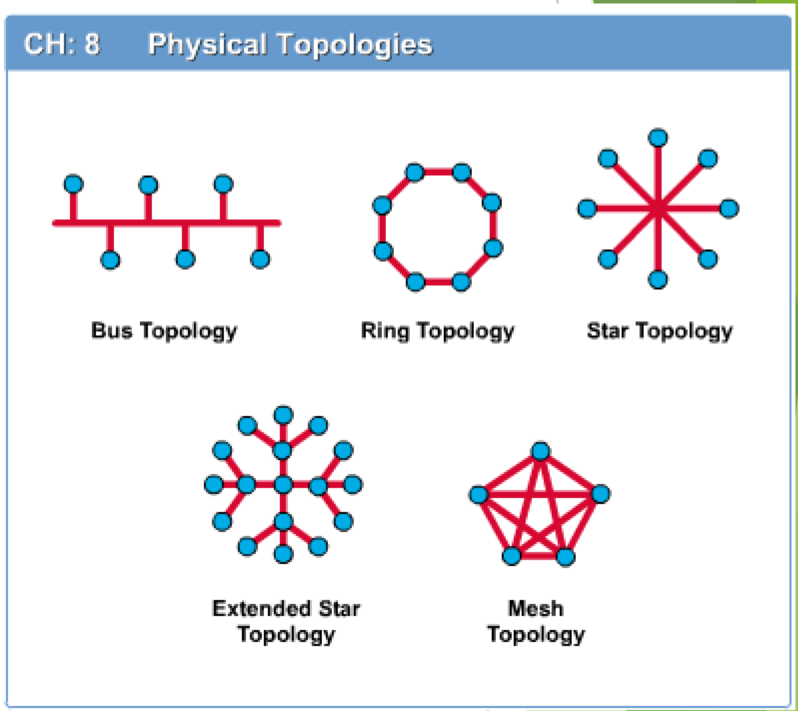
10m	Room	Local Area Network
100m	Building	Local Area Network
1Km	Campus	Local Area Network
10Km	City	Metropolitan Area Network
100Km	Country	Wide Area Network
1000Km	Continent	Wide Area Network
10000Km	Planet	Internet

Applications of Networks

- **Resource Sharing**
  - Hardware (computing resources, disks, printers)
  - Software (application software)
- **Information Sharing**
  - Easy accessibility from anywhere (files, databases)
  - Search Capability (WWW)
- **Communication**
  - Email
  - Message broadcast
- **Remote computing**
- **Distributed processing (GRID Computing)**

Network Topology BSRM

- The network topology defines the way in which computers, printers, and other devices are connected.
- A network topology describes the layout of the wire and devices as well as the paths used by data transmissions.



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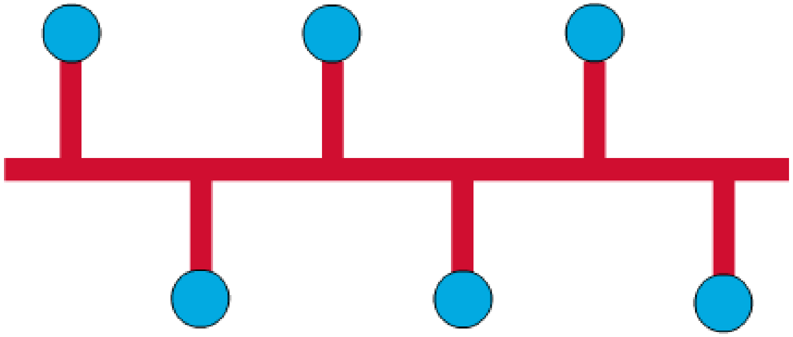
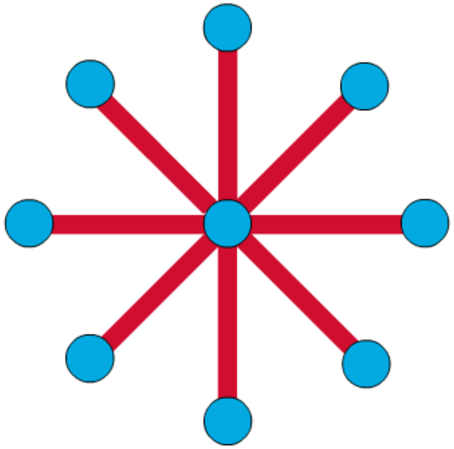
Bus Topology

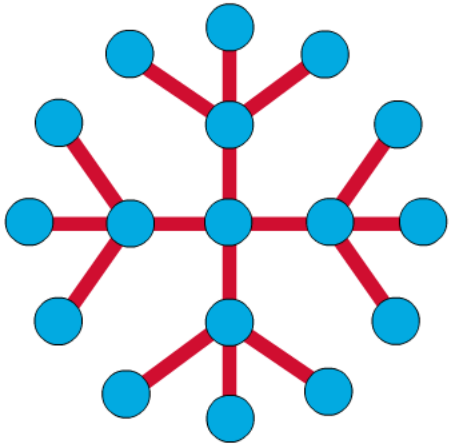
- Commonly referred to as a linear bus, all the devices on a bus topology are connected by one single cable.



Star & Tree Topology

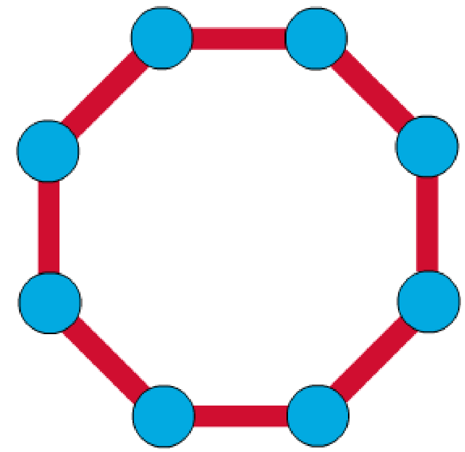
- The star topology is the most commonly used architecture in Ethernet LANs.
- When installed, the star topology resembles spokes in a bicycle wheel.
- Larger networks use the extended star topology also called tree topology. When used with network devices that filter frames or packets, like bridges, switches, and routers, this topology significantly reduces the traffic on the wires by sending packets only to the wires of the destination host.

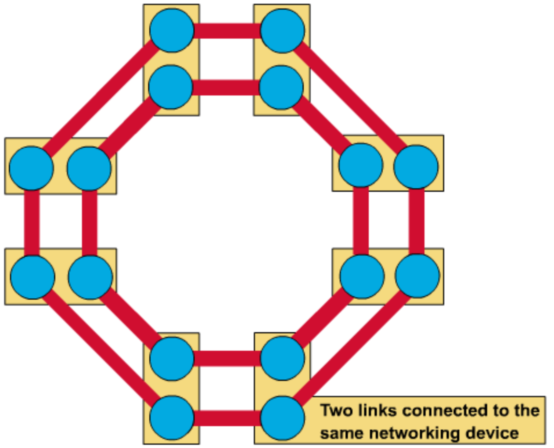




Ring Topology

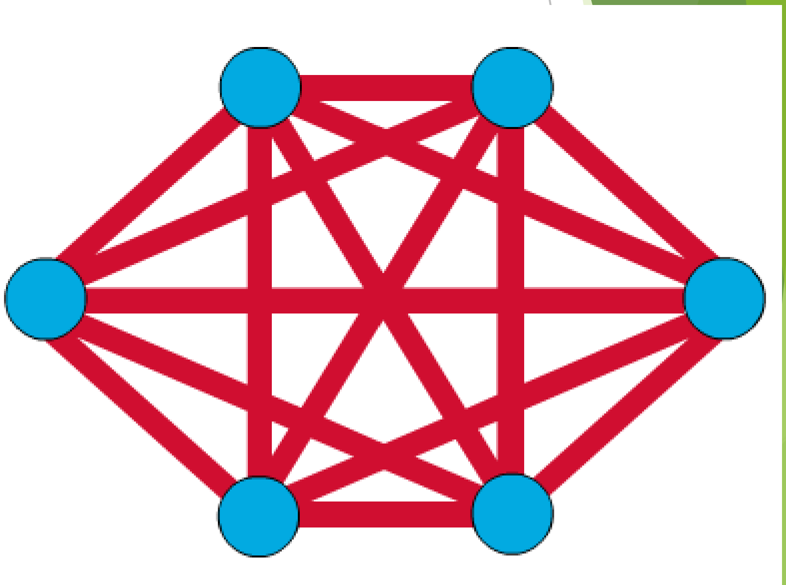
- **A frame travels around the ring, stopping at each node. If a node wants to transmit data, it adds the data as well as the destination address to the frame.**
- **The frame then continues around the ring until it finds the destination node, which takes the data out of the frame.**
  - Single ring – All the devices on the network share a single cable
  - Dual ring – The dual ring topology allows data to be sent in both directions.





Mesh Topology

- The mesh topology connects all devices (nodes) to each other for redundancy and fault tolerance.
- It is used in WANs to interconnect LANs and for mission critical networks like those used by banks and financial institutions.
- Implementing the mesh topology is expensive and difficult.

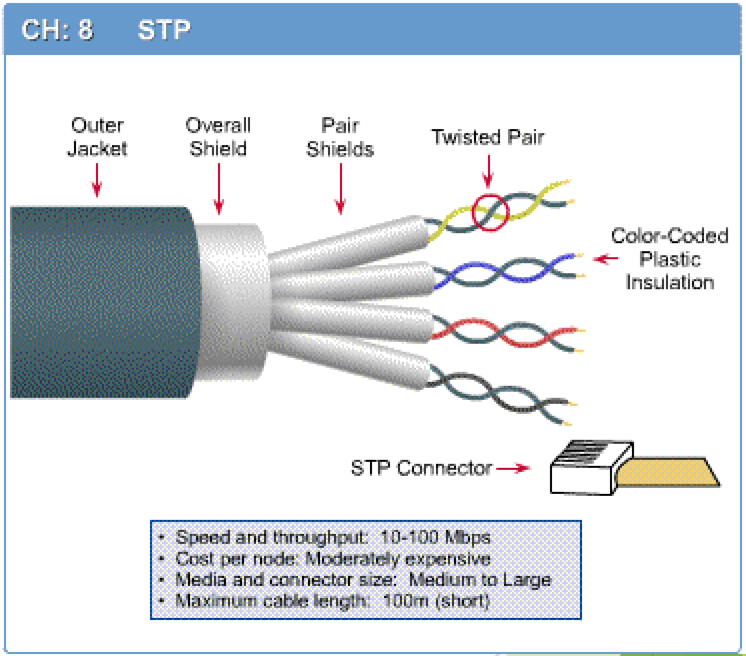


Network Components

- Physical Media
- Interconnecting Devices
- Computers
- Networking Software
- Applications

Networking Media

- Networking media can be defined simply as the means by which signals (data) are sent from one computer to another (either by cable or wireless means).



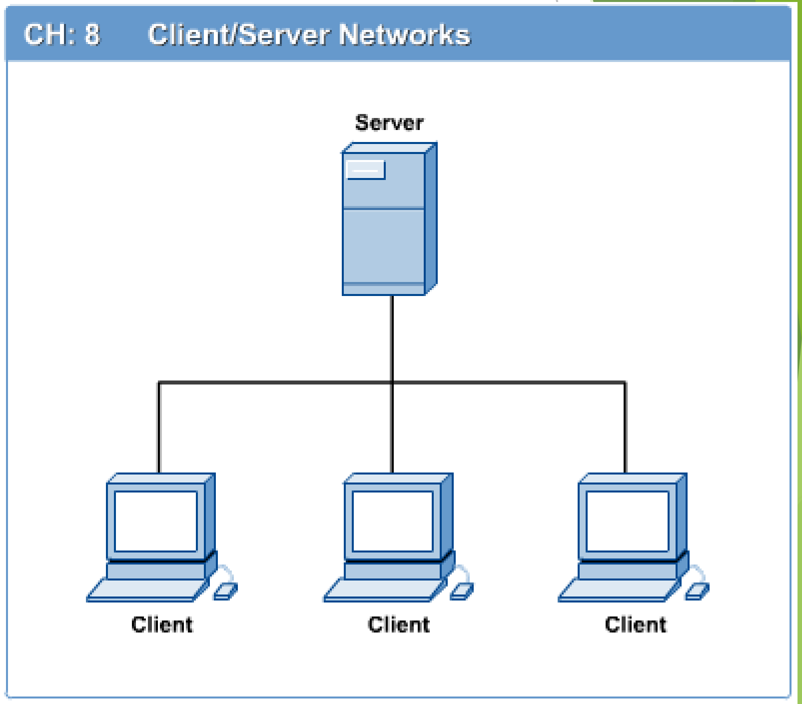
Networking Devices

- HUB, Switches, Routers, Wireless Access Points, Modems etc.

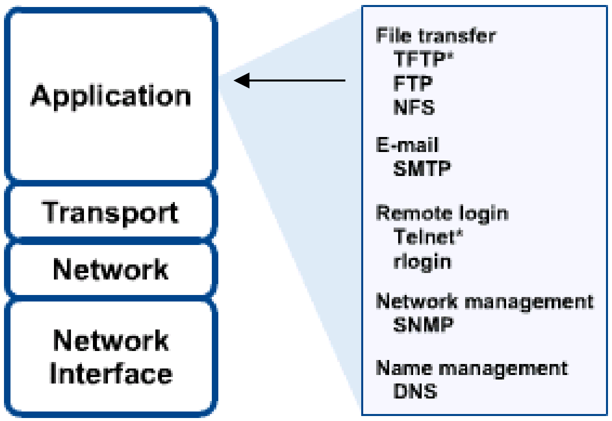
Computers: Clients and Servers

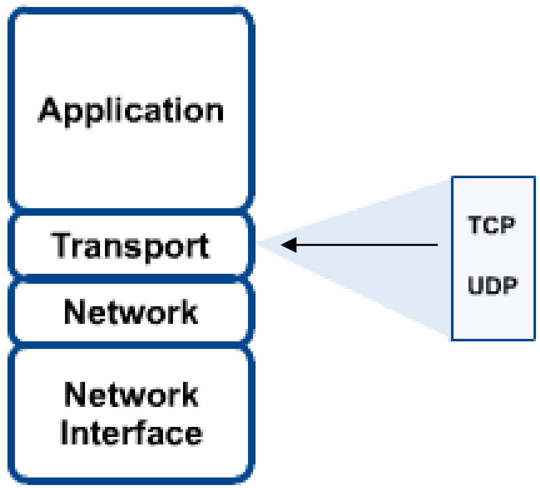
- In a client/server network arrangement, network services are located in a dedicated computer whose only function is to respond to the requests of clients.

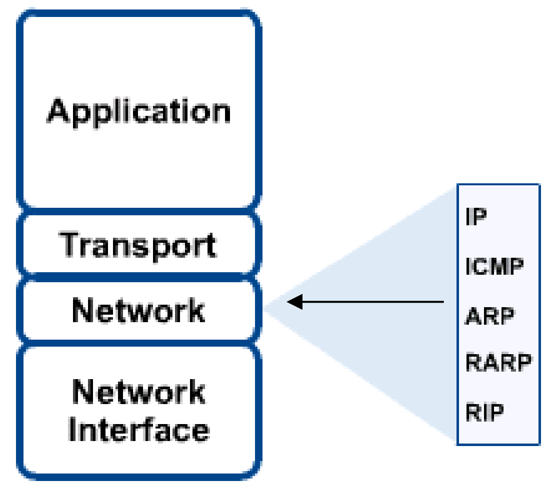
- The server contains the file, print, application, security, and other services in a central computer that is continuously available to respond to client requests.



Networking Protocol: TCP/IP







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Applications

- E-mail
- Searchable Data (Web Sites)
- E-Commerce
- News Groups
- Internet Telephony (VoIP)
- Video Conferencing
- Chat Groups
- Instant Messengers
- Internet Radio

Protocol Hierarchies

- Most networks are organized as a series of layers or levels
  - Each layer offers certain services to the higher layers
- **Protocol**
  - an agreement between the communicating parties on how communication is to proceed.
  - Layer n protocol

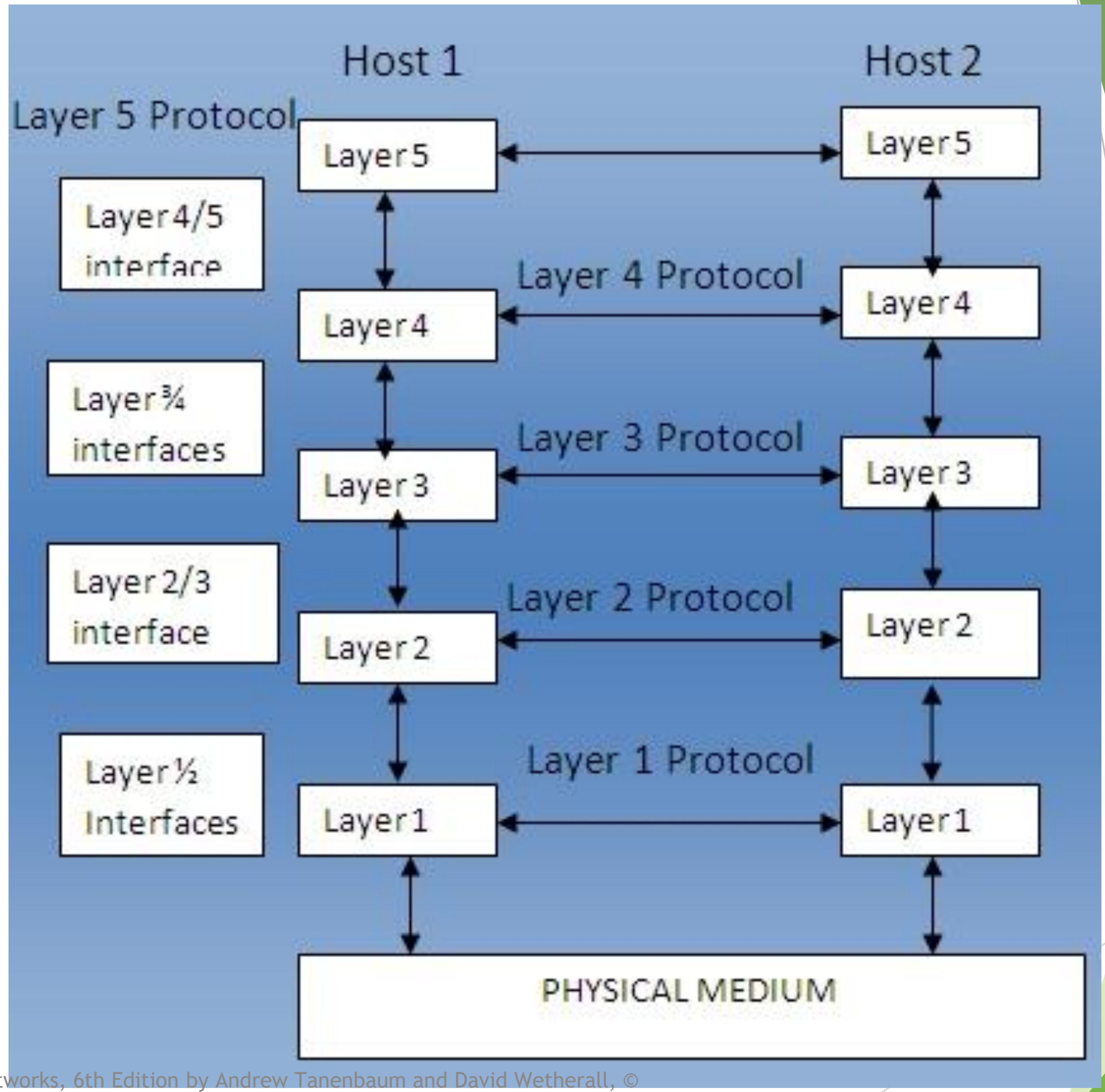


Fig: protocols, layers and interfaces

Peers and Interfaces

- **Peers**

– The entities comprising the corresponding layers on different machines

- **Interface**

– between each pair of adjacent layers

– defines which primitive operations and services the lower layer offers to the upper one.

Entities and Peer Entities

- **An entity**

– an active element in a layer

– can be a software entity (a process), a hardware entity (an I/O chip), or both (an I/O chip with its driver).

- **Peer entities**

– entities in the same layer on different machines

Protocol Stacks

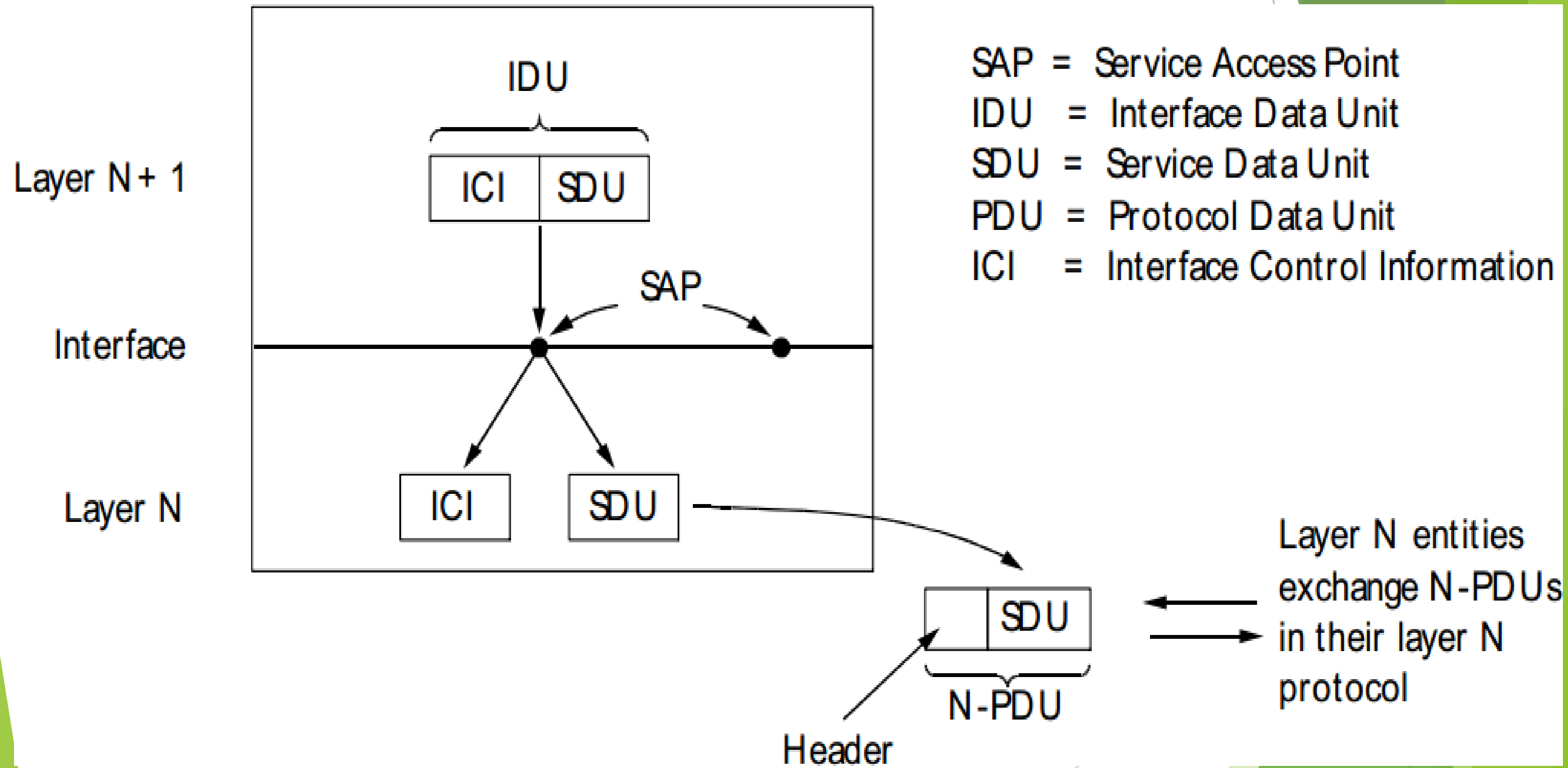
- A protocol stack.

– A list of protocols used by a certain system, one protocol per layer

- Network architecture

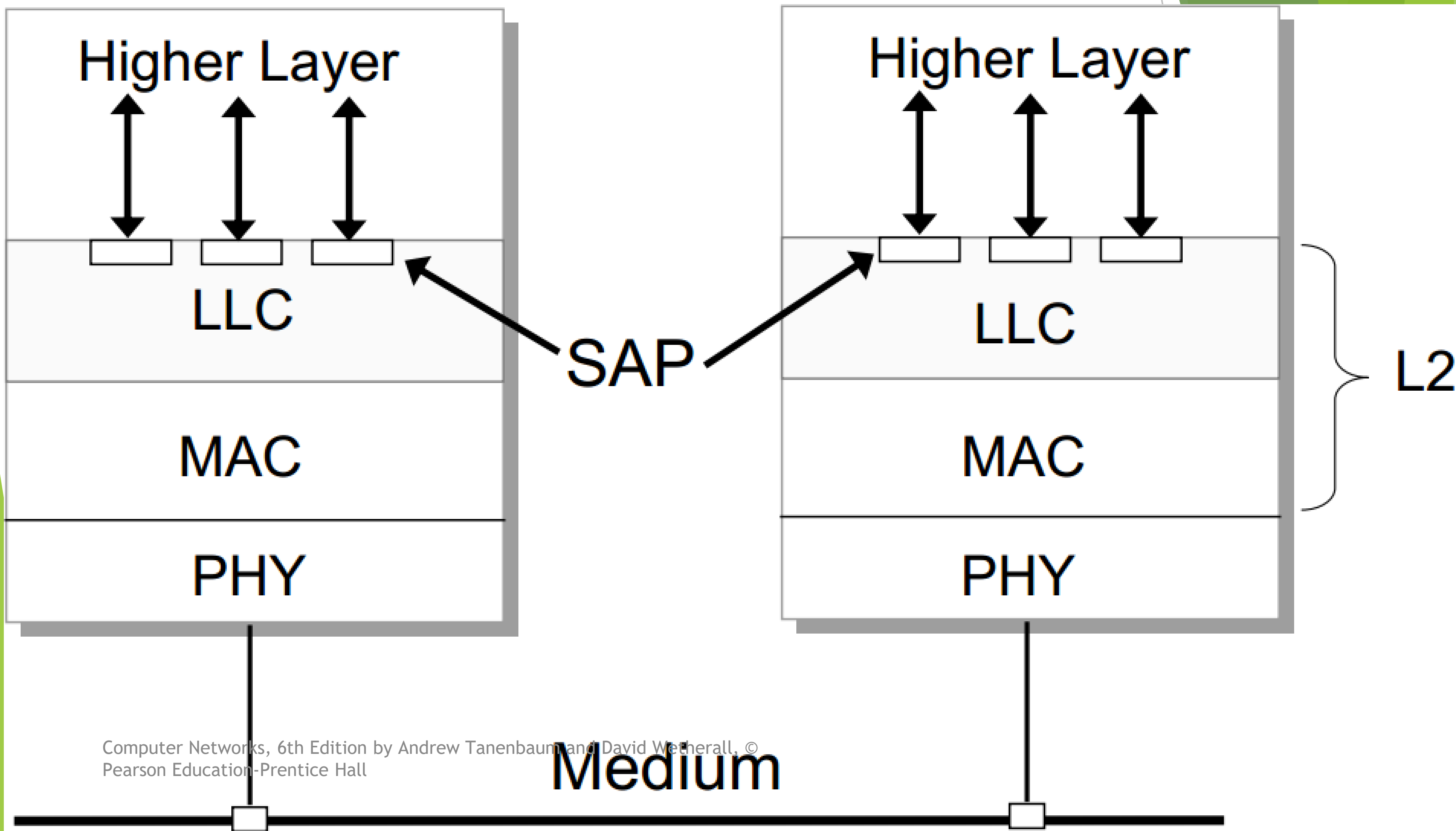
– A set of layers and protocols

Layers at An Interface



SAP = Service Access Point
IDU = Interface Data Unit
SDU = Service Data Unit
PDU = Protocol Data Unit
ICI = Interface Control Information

Example: SAPs in LLC



Till Now

LAN, MAN & WAN
Applications of Networks
Network Topology BSRM
Network Components PINCONA
:protocols, layers and interfaces
Peers, Entities and Peer Entities
Protocol Stacks
Layers at An Interface
Connections Multiplexing

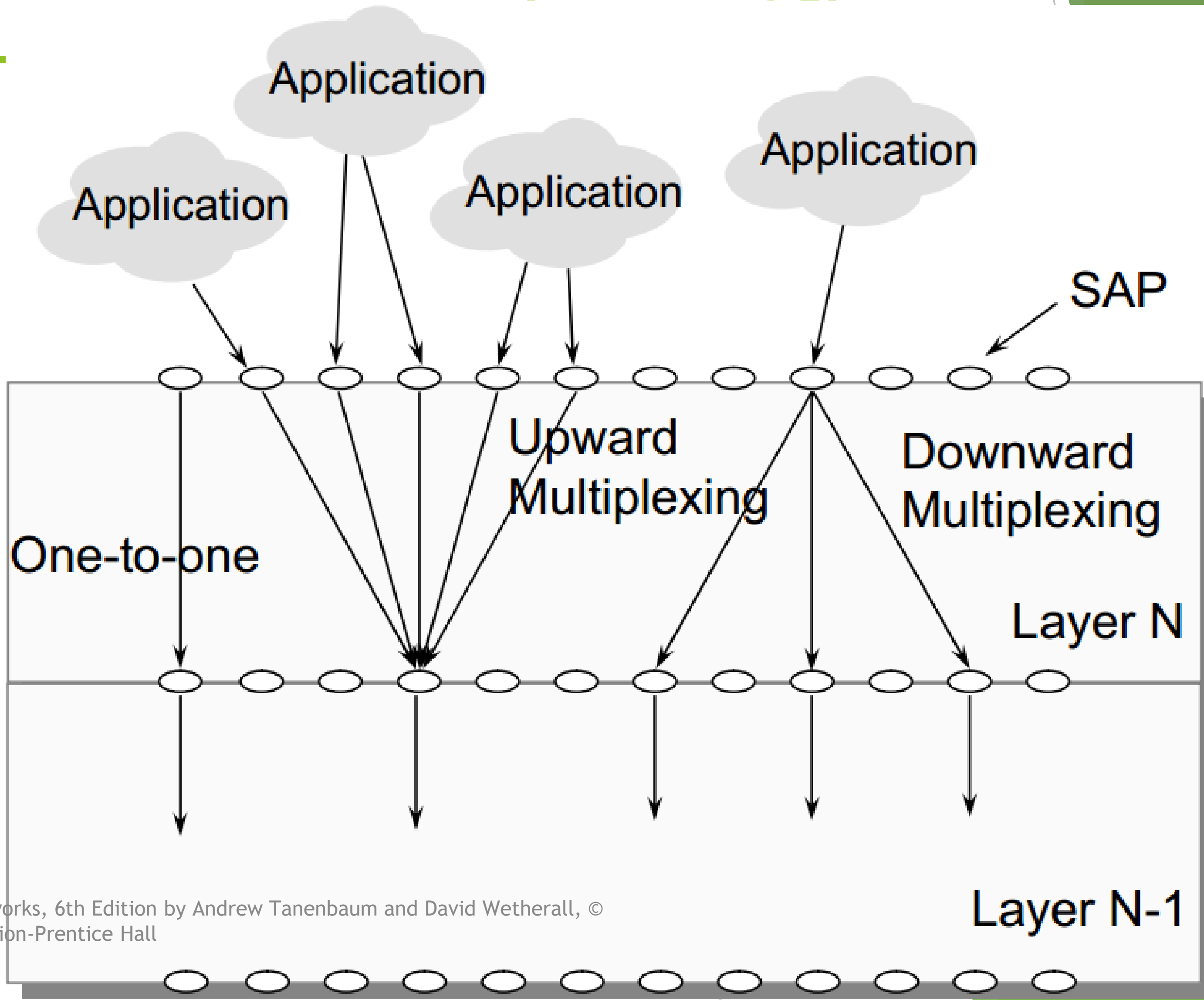
Connections Multiplexing

- Upward Multiplexing: Several layer N connections should be multiplexed into a

layer N-1 connection.

- Downward Multiplexing: a layer N connection uses several layer N-1 connections.

Connections Multiplexing[p-510-



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Connection-Oriented and Connectionless Services

- **connection-oriented**

– Establish a connection first, use the

connection, and then release the connection.

– Example: the telephone system

- **connectionless**

– No connection is established.

– Example: the postal system

Service Primitives

- Request

– layer n+1 to layer n

- Indication

– layer n to layer n+1

– signal an event

- Response

– layer n+1 to layer n

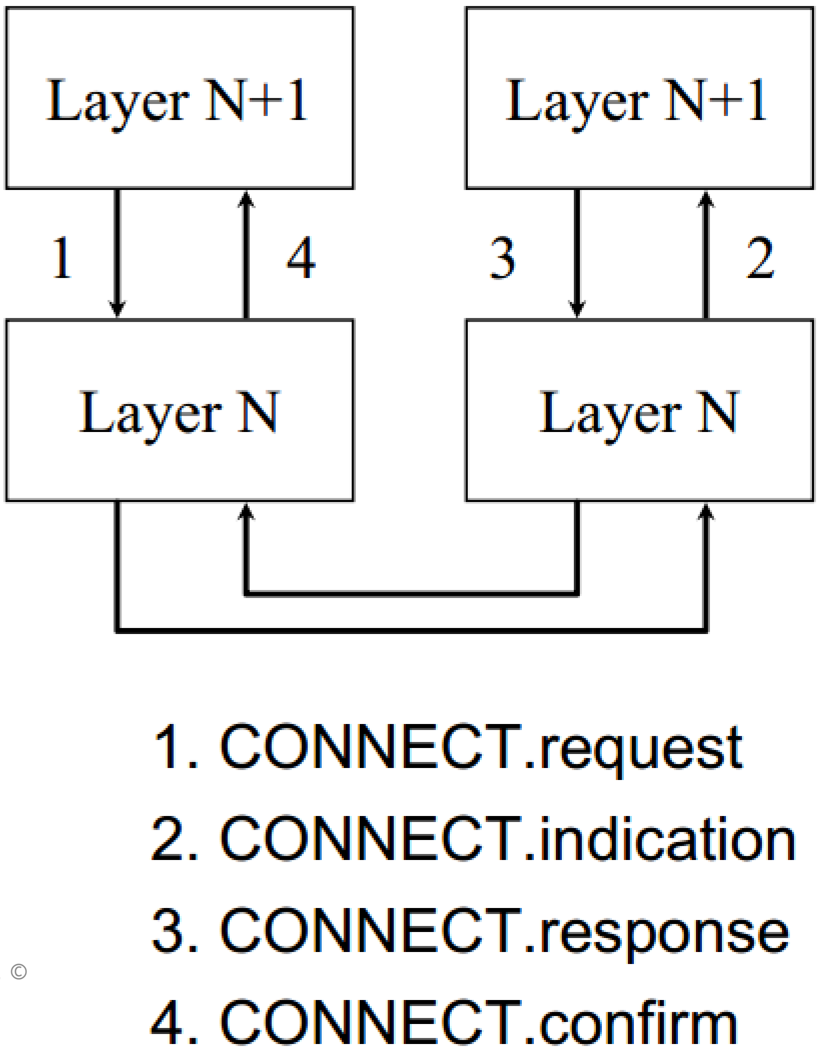
– respond to the previous

indication

- Confirm

– layer n to layer n+1

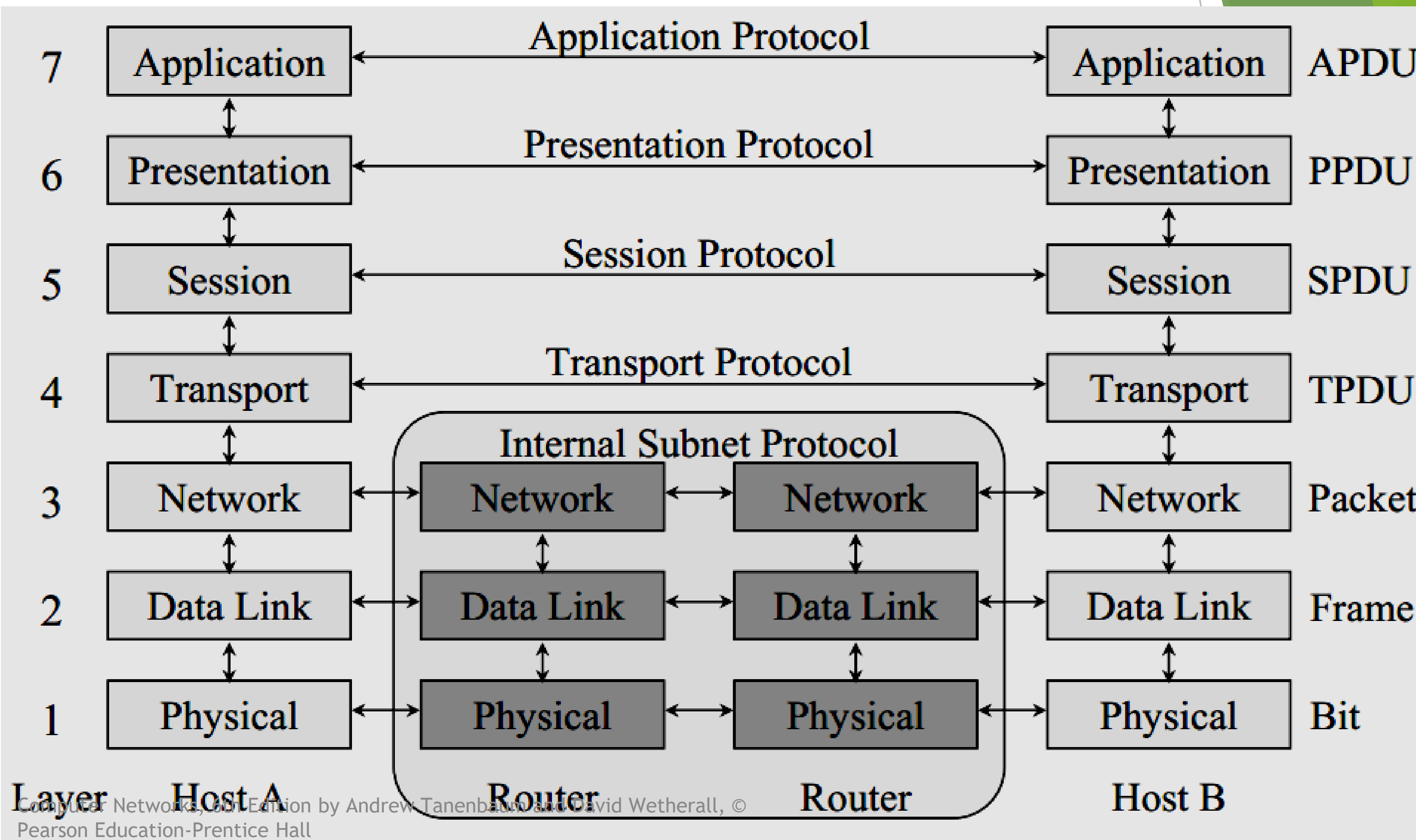
– confirm the previous request



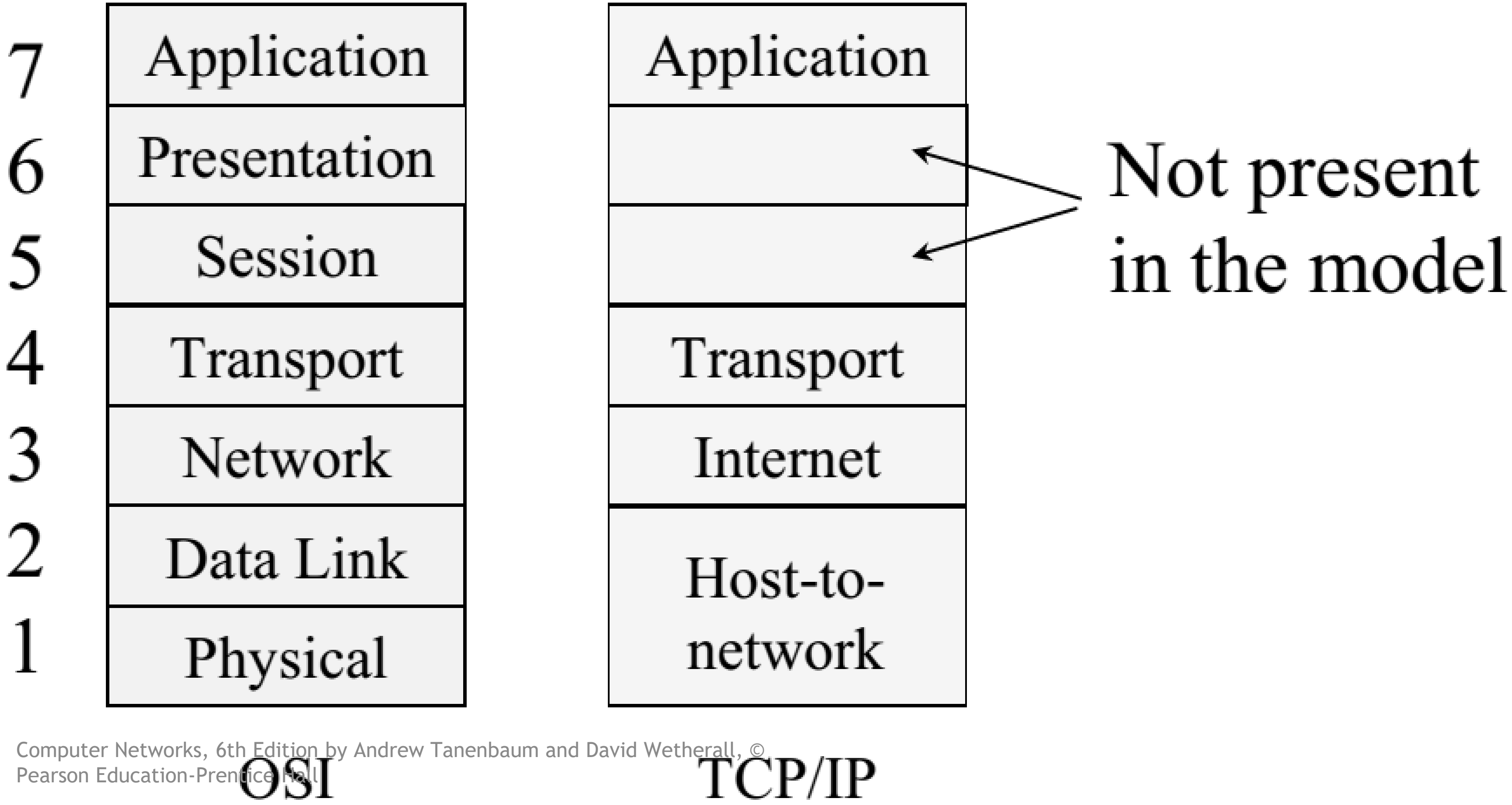
Reference Models

- The OSI Reference Model
  - based on a proposal developed by the International Standards Organization (ISO)
  - a first step toward international standardization of the protocols used in the various layers.
- The TCP/IP Reference Model

The OSI Reference Model



The TCP/IP Reference Model



Physical Layer

- - main function to transmit bits over a communication channel
- - to establish and terminate a connection to a communications medium
- - responsible to make sure that when one side sends a '1' bit the other side will receive '1' bit and not '0' bit.

Data Link Layer

- provides means to transfer data between network entities.
- - it takes the bit streams of data from the Network Layer
- - breaks into frames and passes them to the physical layer.
- - At the receiving end data link layer detects and possibly corrects the errors
- - passes the correct stream to the network layer.
- - also concerned with flow control techniques.

Network Layer

- \- performs network routing,
- \- flow control
- \- error control functions.
- \- prevents the possibility of congestion .

Transport Layer

- -main task to accept data from the Session layer,
- - split them up into smaller units
- - passes them to the Network layer
- - make sure that all the pieces arrived correctly.
- - checks the sequence of packets.

Session Layer

SS

- responsible for controlling exchange information and
- sets up, coordinates, and terminates conversations, exchanges, and dialogs between the applications at each end
- synchronization.
- Provides a session

Presentation Layer

PS

- Responsible to translate different data formats from the representation used inside the computer (ASCII) to the network standard representation and back.
- Computers use different codes for representing character strings so a standard encoding must be used and is handled by the presentation layer.
- Generally in a few words this layer is concerned with the syntax and semantics of the information transmitted.

Application layer

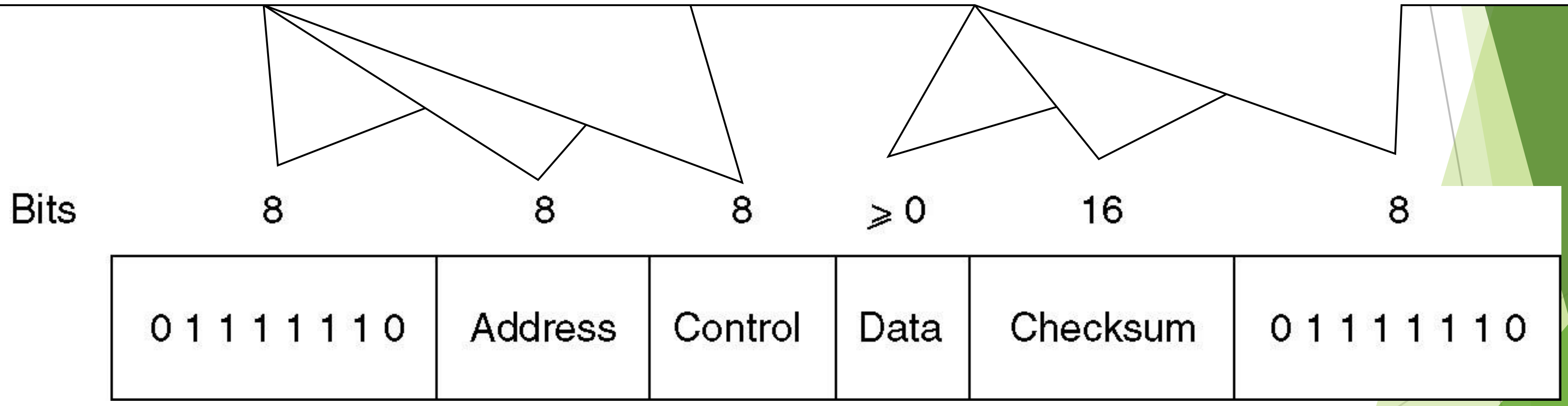
- Makes sure that the other party is identified and can be reached
- If appropriate, authenticates either the message sender or receiver or both
- Makes sure that necessary communication resources exist (for example, is there a modem in the sender's computer?)
- Ensures agreement at both ends about error recovery procedures, data integrity, and privacy
- Determines protocol and data syntax rules at the application level

Example Data Link Protocols

- HDLC – High-Level Data Link Control
- The Data Link Layer in the Internet

High-Level Data Link Control

End of frame (same as start of frame)



Frame format for bit-oriented protocols

High-Level Data Link Control (2)

Bits	1	3		1	3
(a)	0	Seq		P/F	Next
(b)	1	0	Type	P/F	Next
(c)	1	1	Type	P/F	Modifier

Control field of

(a) An information frame.

(b) A supervisory frame.

(c) An unnumbered frame.

Information Frame

- SEQ – 3 bits sequence numbers are used
- NEXT – used for piggybacking acknowledge
  - All of the HDLC protocols adhere to the convention that instead of sending the sequence number for the frame that has been received correctly, the acknowledgement contains the sequence number of the next frame that is expected (not received yet).

Information Frame

- The P/F bit stands for Pool/Final and it is used when a computer is polling a group of terminals.
  - When used as P, the computer is inviting the terminals to send data. All the frames from the terminal, except the final one, have the P/F flag set to P. The final one is F.
  - The F flag is sometimes used to force the other machine to send a supervisory frame immediately, without waiting for reverse traffic where to piggyback the window information.

Supervisory Frame

- Various kinds of SUPERVISORY frames are distinguished by the type filed:
  - **[00]Type 0 is an acknowledgement frame** (called RECEIVE READY), used to indicate the next frame expected
  - **[01]Type 1 is a negative acknowledgement frame** (called REJECT), used to indicate an error. The NEXT field indicates the first frame in sequence that has not been received correctly. The sender is requested to send all frames beginning at NEXT

Supervisory Frame

- **[10]Type 2 is RECEIVE NOT READY**. It acknowledges all frames up to but not including the NEXT one. It also tells the sender to stop sending. When the receiver is ready again, it sends a RECEIVE READY.
- **[11]Type 3 is a SELECTIVE REJECT**. It calls for retransmission of only the frame specified. Thus, if a receiver wants to buffer out of sequence frames, it can force the retransmission of a specific frame by using a selective reject.

Unnumbered Frame

- UNNUMBERED FRAME is used for control and to carry data when **unreliable connectionless** service is required.
- Using this frame, a number of commands can be sent:
  - **DISConect** – allows a machine to announce its intention to bring down the connection
  - **SABM** (Set Asynchronous Balanced Mode) – resets the line
  - **FRMR** (Frame Reject) – indicates that a frame with correct checksum but incorrect semantics has arrived
- More control frames are actually available for initialization, polling and status reporting.